Muhammad Mahasin Irfan

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Research Interests

Human centered security and privacy; trustworthy AI measurement; machine learning systems; efficient ML inference.

Education

Lahore University of Management Sciences (LUMS) Lahore, Pakistan

B.S. in Computer Science; CGPA: 3.85/4.00 *September 2023 – August 2027*

- Dean's Honor List in every completed academic year (2023–24, 2024–25, 2025–26).

Workshop Paper

M. M. Irfan, M. S. Hassan, and F. Zaffar. "Decomposing Error in Continual World Models." **NeurIPS 2026 Workshop on Continual World Models**, Idea Track. *Accepted*. PDF.

Selected Research Experience

Security and Privacy in Mobile Phone Shops June 2026 – Present

Joint First Author Researcher, LUMS/CIRCLE *Lahore, Pakistan*

Advisors/Collaborators: Dr. Mobin Javed; Dr. Katharina Krombholz (CISPA); Dr. Lea Grober (ICSI Berkeley; formerly CISPA)

- Co-designed and led participant-facing fieldwork for an IRB-approved study of how mobile-phone shopkeepers act as security and privacy intermediaries for low-income communities in Pakistan.
- Conducted 15 semi-structured, approximately 30-minute Urdu interviews about authentication mental models, passwords and device PINs, account recovery, data transfer, privacy advice, and fraud.
- Built participant rapport; transcribed, translated, and coded the interviews; and contributed to collaborative reflexive thematic analysis for a joint first author manuscript in preparation.

Evaluation of Continual World-Model Updates July 2026 – Present

First Author Researcher, LUMS *Lahore, Pakistan*

Advisors/Collaborators: Dr. Fareed Zaffar

- Derived an exact decomposition of post-update prediction error into inherited error and update mismatch, showing why endpoint accuracy alone can obscure whether an adaptive model faithfully tracked an environmental change.
- Demonstrated opposite endpoint/mismatch rankings in 544/544 controlled one-step trials and reproduced the phenomenon across 500 learned-model runs on a nonlinear damped-pendulum simulator.
- Developed the formulation, implemented and analyzed the experiments, and wrote the complete first draft of the accepted workshop paper.

Optimal ML Workload Splitting Across Heterogeneous Memory Controllers November 2025 – Present

Undergraduate Researcher, LUMS *Lahore, Pakistan*

Advisors/Collaborators: Dr. Fareed Zaffar; Humza Ikram (CMU); Dr. Hashim Sharif (AMD Research)

- Developing layer-partitioning methods for heterogeneous GPU and processing-in-memory execution, evaluated on four CNNs and four LLMs ranging from MobileNetV2 to Qwen3-32B.
- Extended a Newton PIM model in Ramulator using C++ to explore additional bank groups and larger row sizes with corresponding t_{FAW} adjustments, improving simulated throughput by 1.3× and 1.5× over the original configuration.
- Proposed a prediction-guided search heuristic that evaluates up to 7–8× fewer split candidates while selecting configurations within 1% of exhaustive-oracle end-to-end latency in the best evaluated cases; systems manuscript in preparation.

Research Internship

AI Optimization Intern June 2025 – August 2025
10x Engineers *Lahore, Pakistan*
Advisors/Collaborators: Dr. Saad bin Nasir; Dr. Khurram Usman Mazher

- Extended QuantX for simulated post-training quantization of diffusion pipelines, adding component-aware traversal and configurable linear/convolutional weight and activation granularities for Stable Diffusion 1.5 and SDXL, with exploratory SD3.5 support.
- Built diffusion-trajectory calibration and activation-statistics infrastructure and implemented SmoothQuant-style equivalent scaling; studied PTQ4DM, Q-Diffusion, ViDiT-Q, and SVDQuant to guide the design.
- Diagnosed catastrophic W4 degradation: excluding the convolution path reduced LPIPS from 0.5713 to 0.3380, while subsequent smoothing improved it to 0.3357 under the recorded evaluation protocol.
- Investigated efficient on-device generation, where keeping inference local can reduce the need to transmit sensitive prompts or generated content to a remote service. [Project story]

Teaching Experience

Teaching Assistant, Computer Systems and Assembly Language (CS225) Fall 2026
LUMS; instructors: Dr. Zartash Afzal Uzmi and Dr. Jahangir *~350 students, two sections*

- Develop quizzes, assignments, and a new systems laboratory; grade examinations and conduct tutorials.

Teaching Assistant, Foundations of Data Science Spring 2026
LUMS; instructor: Dr. Mobin Javed *~200 students*

- Designed Python and data-visualization assignments, automated tests, and Jupyter/Otter-Grader workflows; held office hours and supported course projects.

Teaching Assistant, Introduction to Programming (CS100) Spring 2026
LUMS; instructor: Dr. Fareed Zaffar *~180 students*

- Created assignments, held programming labs and a tutorial, and provided student-facing office hours.

Academic Service

Selected Reviewer, NeurIPS 2026 Workshop on Continual World Models 2026
Selected by the workshop Technical Program Committee

- Reviewed one anonymous workshop submission and submitted a written review and recommendation to the program committee.

Technical and Research Skills

Programming/tools: Python, C++, C, PyTorch, TVM, TensorRT, Ramulator, Git, Linux, Docker, Jupyter, LaTeX.
ML systems/evaluation: Model evaluation, quantization, performance benchmarking, autotuning, heterogeneous execution, memory-system simulation.
Human studies: Randomized controlled trials, semi-structured interviewing, usability studies, reflexive thematic analysis; Urdu and Punjabi fieldwork.

Selected Honors

Dean's Honor List (all completed years) | National Mathematics Talent Contest, Pakistan top 20 | SEAMO Gold Medal, third in Pakistan | Purple Comet! Math Meet, first-place team in Pakistan